Jerry Quinn

Senior Data Platform & MLOps Engineer | Big Data & Machine Learning Pipeline Specialit

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SUMMARY

Senior Data Platform & Cloud Data Engineer with 9+ years of experience designing, building, and operating enterprise-scale data platforms that power analytics, business intelligence, real-time insights, and production machine learning systems. Proven ability to translate complex business objectives—including customer personalization, financial accuracy, healthcare outcomes, and operational efficiency—into scalable, secure, and compliant data architectures.

Expertise spans modern cloud data ecosystems, including data warehouses and lakehouse architectures, ELT/ETL frameworks, streaming pipelines, and analytics engineering. Highly proficient in Python, SQL, Spark, and modern orchestration technologies for building reliable and high-performance data pipelines.

Recognized for leading legacy data platform modernization, implementing robust data quality and governance frameworks, and establishing CI/CD-driven DataOps and MLOps practices that accelerate delivery while improving system reliability and scalability. A collaborative partner to business stakeholders, analytics teams, and data scientists, consistently delivering trusted, analytics-ready data that enables informed decision-making and measurable business impact.

TECHNICAL SKILLS

- **Data Engineering & Data Management**
  Data Processing, Data Management, Data Integration, Data Modeling, Data Warehousing, ETL/ELT Development, Data Pipelines, Data Streaming, Data Analytics, Data Lineage, Data Governance, Data Quality, Data Architecture
- **Programming & Query Languages**
  Python, SQL, Scala, Java, PySpark, T-SQL, PL/SQL, Shell Scripting, R
- **Big Data & Distributed Processing**
  Apache Spark, Spark Structured Streaming, Hadoop, Hive, HiveQL, Apache Flink, Apache Kafka, Apache NiFi, Pig, HBase
- **Cloud Platforms**
  Amazon Web Services (AWS), Google Cloud Platform (GCP), Microsoft Azure
- **AWS Data Services**
  S3, Redshift, Glue, Lambda, Kinesis, EMR, Athena, DynamoDB, Step Functions
- **Google Cloud Data Services**
  BigQuery, Dataflow, Pub/Sub, Dataproc, Cloud Composer
- **Azure Data Services**
  Azure Synapse Analytics, Azure Data Factory
- **Data Warehouses & Databases**
  Snowflake, PostgreSQL, Cassandra, MongoDB, Oracle, SQL Server, MySQL, DB2, Sybase, Teradata
- **Data Lake & Lakehouse Platforms**
  Databricks, Delta Lake, CDP (Cloudera Data Platform)
- **Data Integration & ETL Tools**
  Airflow, dbt, Fivetran, Informatica, Talend, Pentaho, DataStage, SSIS, RoundhousE, Luigi
- **Infrastructure & DevOps**
  Terraform, CloudFormation, Docker, Kubernetes, Ansible, Jenkins, GitHub Actions, Git
- **Monitoring & Observability**
  Datadog
- **Business Intelligence & Visualization**
  Tableau, Power BI, Qlik, Looker, MicroStrategy, Cognos, SSRS
- **Data Architecture & Modeling**
  Star Schema, OLAP, OLTP, Data Marts, OLAP Cubes
- **Security, Compliance & Governance**
  RBAC, Data Encryption, PCI DSS, SOX, GDPR, HIPAA

PROFESSIONAL EXPERIENCE

Senior Data Platform Engineer

JPMorganChase | Aug 2020 - Jan 2026

- Architected and operated enterprise-scale cloud data platforms leveraging DataOps, DevOps, and MLOps best practices, supporting mission-critical analytics, business intelligence, real-time data processing, and production machine learning workloads across large financial datasets.
- Automated infrastructure provisioning and environment management using Terraform, while containerizing data services with Docker and Kubernetes to enable scalable, secure, and highly available data platform environments aligned with enterprise governance and compliance standards.
- Integrated Fivetran-managed ingestion pipelines with dbt and dbt Fusion, standardizing downstream ELT transformations, implementing automated data quality checks, and enabling scalable analytics engineering workflows across multiple business domains.
- Deployed and managed self-hosted Airbyte on Docker and Kubernetes, enabling customizable data connectors, improved pipeline scalability, and greater control over sensitive financial data movement between operational systems and analytics platforms.
- Designed and maintained robust CI/CD pipelines using Jenkins, GitHub Actions, GitLab CI, and Azure Pipelines, supporting automated testing, version control, and deployment for data engineering, analytics, and ML workflows written in Python, PySpark, SQL, and dbt.
- Administered and optimized the Databricks Lakehouse platform utilizing Apache Spark and Delta Lake, enabling unified batch and streaming data processing while integrating with enterprise data warehouses including Snowflake, BigQuery, Amazon Redshift, and AWS Athena.
- Implemented modern ELT and analytics engineering frameworks using dbt and dbt Fusion, creating modular, reusable transformation layers, enforcing data quality rules, and accelerating delivery of trusted datasets for analysts, data scientists, and business stakeholders.
- Built and supported real-time, event-driven data pipelines using Apache Kafka, enabling low-latency ingestion, scalable streaming analytics, and near real-time data availability for operational dashboards, risk monitoring, and downstream ML applications.
- Led SQL Server analytics and BI modernization initiatives, developing complex T-SQL queries, stored procedures, and optimized data models, while migrating legacy IBM Cognos BI objects to SSRS reports, requiring extensive data transformations and performance tuning.
- Enabled and supported end-to-end MLOps pipelines, facilitating feature engineering, model training, automated deployment, and model monitoring while leveraging Datadog observability tools and serving as a senior escalation point for production data platform, streaming, and ML pipeline issues.

Data Platform Engineer

Red Pill Analytics | Aug 2018 - Aug 2020

- Collaborated closely with business, product, marketing, and analytics stakeholders to gather detailed requirements and translate complex marketing and AI use cases into scalable technical solutions, implementing privacy-aware data engineering practices that supported customer 360 initiatives, advanced personalization strategies, and enterprise analytics programs.
- Designed and maintained end-to-end data pipelines and centralized feature stores supporting machine learning workflows, enabling AI-driven personalization, predictive modeling, and privacy-centric customer identity resolution across paid, owned, and earned marketing channels, improving the consistency and usability of ML features across multiple downstream systems.
- Built and automated scalable ETL/ELT frameworks using SQL Server, SSIS, T-SQL, PostgreSQL, BIML Express, and SQL Server Agent, ingesting, transforming, validating, and orchestrating large volumes of structured and semi-structured datasets to support model training, batch inference pipelines, and enterprise reporting workloads.
- Developed and maintained enterprise data warehouses and subject-oriented data marts spanning Marketing, Sales, Product, HR, and Customer Engagement domains, implementing dimensional modeling, slowly changing dimensions (SCD), indexing, and partitioning strategies to support scalable analytics, reporting, and ML feature engineering.
- Partnered with data scientists, ML engineers, and analytics teams to operationalize machine learning models by ensuring high-quality, reproducible datasets and reliable feature pipelines, enabling production ML scoring for audience segmentation, churn prediction, attribution modeling, and personalized marketing campaigns.

- Implemented comprehensive data quality, validation, and monitoring frameworks using SQL Server and PostgreSQL, creating automated checks, anomaly detection rules, and data validation pipelines that significantly improved feature reliability and ensured consistent inputs for ML models and marketing decision systems.
- Built and maintained mission-critical ML scoring pipelines and data streams, integrating outputs with marketing automation, email marketing, loyalty programs, digital media platforms, and personalization engines to deliver real-time insights, customer recommendations, and targeted engagement strategies.
- Led the development of reusable ETL frameworks and database CI/CD pipelines using BIML Express and RoundhousE, standardizing schema versioning, automated deployments, and environment consistency across development, staging, and production environments for data engineering workflows.
- Optimized large-scale SQL Server and PostgreSQL workloads by implementing indexing strategies, query tuning, incremental data loading patterns, and table partitioning, significantly improving ETL performance, reducing data refresh times, and accelerating machine learning model training cycles.
- Collaborated continuously with business, product, and analytics leadership to align data engineering architectures with evolving marketing and AI objectives, designing privacy-compliant data pipelines, scalable data models, and ML-ready datasets that enabled deeper customer insights, improved personalization capabilities, and data-driven decision making.

Cloud Data Engineer

Red Pill Analytics | Jan 2018 - Jul 2018

- Architected and delivered enterprise-scale data engineering solutions supporting Empower’s retirement, recordkeeping, and wealth management platforms, enabling reliable data processing and analytics for 19M+ participants and 88K+ organizations, while maintaining high standards for data accuracy, performance, and regulatory compliance.
- Designed and maintained robust ETL and ELT data pipelines to ingest, cleanse, standardize, and transform large volumes of financial, transactional, and customer data sourced from relational databases including Oracle, SQL Server, and PostgreSQL, as well as third-party financial data providers and internal operational systems.
- Played a key role in early cloud data platform modernization initiatives, leveraging services across AWS, Microsoft Azure, and Google Cloud Platform to gradually transition legacy on-premise data infrastructure toward scalable, resilient, and cloud-first analytics architectures capable of supporting growing data volumes and advanced analytics use cases.
- Implemented and managed cloud-based data warehouses and analytical layers using Snowflake and Google BigQuery, enabling high-performance analytics and self-service reporting capabilities for actuarial teams, compliance officers, and business intelligence stakeholders.
- Developed optimized ELT workflows that strategically pushed complex data transformations closer to the data warehouse layer, improving query execution performance, reducing data latency for downstream consumers, and lowering infrastructure and operational costs associated with traditional ETL processing.
- Built and optimized dimensional and analytical data models including fact and dimension tables designed to support retirement plan analytics, contribution tracking, investment performance analysis, participant behavior insights, and regulatory financial reporting.
- Applied robust business logic, financial rules, and validation checks across multiple data pipelines to ensure the accuracy and consistency of critical financial metrics such as account balances, vesting schedules, employee contributions, distributions, and employer match calculations.
- Collaborated closely with product managers, compliance teams, finance stakeholders, and business analysts to translate complex domain requirements into trusted, analytics-ready datasets that powered executive dashboards, regulatory reporting workflows, and customer-facing insights across retirement and wealth management platforms.
- Implemented data governance, security, and compliance best practices, including role-based access control (RBAC), encryption for data at rest and in transit, auditing mechanisms, and controlled data access policies to meet stringent financial services regulatory standards.
- Automated recurring data workflows and operational processes using Python and advanced SQL scripting, reducing manual data processing, minimizing operational risk, improving pipeline reliability, and enabling faster delivery of trusted financial datasets to downstream analytics and reporting systems.

Clinical Data Engineer

Red Pill Analytics | Oct 2015 - Dec 2017

- Designed, developed, and maintained scalable ETL data pipelines supporting healthcare operations, clinical data integration, and care-coordination workflows, ensuring reliable data availability for analytics, reporting, and operational decision-making across multiple healthcare departments.

- Built both batch and near-real-time data ingestion pipelines to integrate large volumes of data from electronic health record (EHR) systems, claims processing feeds, care management platforms, pharmacy systems, and third-party healthcare vendors, enabling unified access to patient and operational data.
- Implemented robust data integration and transformation processes to consolidate clinical, operational, and financial datasets into centralized enterprise data warehouses and early-stage data lake architectures, improving data accessibility and enabling advanced healthcare analytics.
- Ensured strong data quality, validation, and reconciliation processes across pipelines handling sensitive PHI and HIPAA-regulated healthcare data, implementing automated checks, data standardization rules, and anomaly detection to maintain accuracy and compliance.
- Optimized pipeline performance, reliability, and data freshness by implementing monitoring, logging, alerting, and automated failure recovery mechanisms, reducing pipeline downtime and ensuring consistent delivery of time-sensitive healthcare data.
- Collaborated closely with clinicians, healthcare operations teams, analytics groups, and care-coordination stakeholders to translate complex healthcare workflows and reporting requirements into production-ready data pipelines and analytics-ready datasets.
- Supported healthcare analytics, regulatory reporting, and population health management initiatives by delivering clean, standardized, and well-modeled datasets that enabled analysts and care teams to generate insights into patient outcomes, treatment effectiveness, and operational performance.
- Implemented data governance practices and metadata documentation standards, improving data lineage visibility, traceability of healthcare datasets, and maintainability of ETL pipelines across evolving clinical and operational systems.
- Assisted in standardizing ETL frameworks, data modeling practices, and reusable pipeline components, improving development efficiency, pipeline scalability, and long-term maintainability of healthcare data infrastructure.
- Contributed to compliance-aligned data engineering practices supporting HIPAA requirements, healthcare reporting standards, and internal audit processes, ensuring secure handling, storage, and access control of sensitive patient information across the data platform.

EDUCATION

2011 - 2015
Gaffney, SC, United States

Bachelor of Science in Computer Science
Limestone University